Advancing the Calibration-Compute Frontier of LLM Classifiers via Asymmetric Duos

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Abstract

The importance of reliable uncertainty estimation in large language models (LLMs) has become increasingly pronounced, yet modern models remain poorly calibrated and frequently overconfident. In this study, we address this by proposing an efficient method for LLM classifier calibration using the *Asymmetric Duos* framework. Our approach combines a strong base model with a smaller sidekick model via logit-space fusion, utilizing global scaling parameters optimized on a validation set. Across diverse benchmarks, this Asymmetric LLM Duo consistently achieves best or second-best performance on expected calibration error and negative log-likelihood, without degrading predictive accuracy. More importantly, these gains are realized at a fraction of the computational cost of more complex baselines, placing our method on the calibration-compute Pareto frontier. We further demonstrate that our method acts orthogonal to various established calibration-aware techniques, meaning it can be easily combined with them to consistently boost their performance. Finally, our analysis of the logit-space behavior illustrates how asymmetric fusion successfully mitigates base model overconfidence. Overall, our findings demonstrate that the benefits of an asymmetric fusion successfully transfer to the language domain, providing a simple, highly cost-efficient approach to improving uncertainty estimation in LLM classifiers.

1 Introduction

Large language models (LLMs) have demonstrated remarkable predictive performance across a wide range of tasks and domains [Brown et al., 2020, OpenAI, 2023, Touvron et al., 2023, Grattafiori et al., 2024, Minaee et al., 2024]. These capabilities can be further increased for downstream tasks through fine-tuning methods [Hu et al., 2022, Yang et al., 2024a, Wang et al., 2024, Lialin et al., 2024, Wang et al., 2022]. Despite the strong predictive performance of LLMs, these models often remain poorly calibrated and exhibit overconfidence [Guo et al., 2017, Xiong et al., 2024, Leng et al., 2025, Shorinwa et al., 2024], and fine-tuning procedures can further amplify this behavior [Jiang et al., 2020]. LLMs being overconfident in their predictions can yield serious problems for critical decision-making domains [Kumar et al., 2023], and being overconfident can also undermine the reliability and restrict the use cases of LLMs in those domains [Li et al., 2025]. Therefore, in real-world applications, these neural networks should also be able to indicate when they can be incorrect [Guo et al., 2017]. This capability is known as uncertainty quantification or confidence calibration.

Overconfidence is not specific to LLMs. Modern neural networks are often poorly calibrated and exhibit overconfidence [Guo et al., 2017]. Several works in the literature have attempted to mitigate this problem. One such solution is the recently proposed *Asymmetric Duo* [Zhou et al., 2025]. They showed that using a smaller sidekick neural network model together with a larger base model can improve the uncertainty estimates of those neural networks. Their results suggest that uncertainty

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

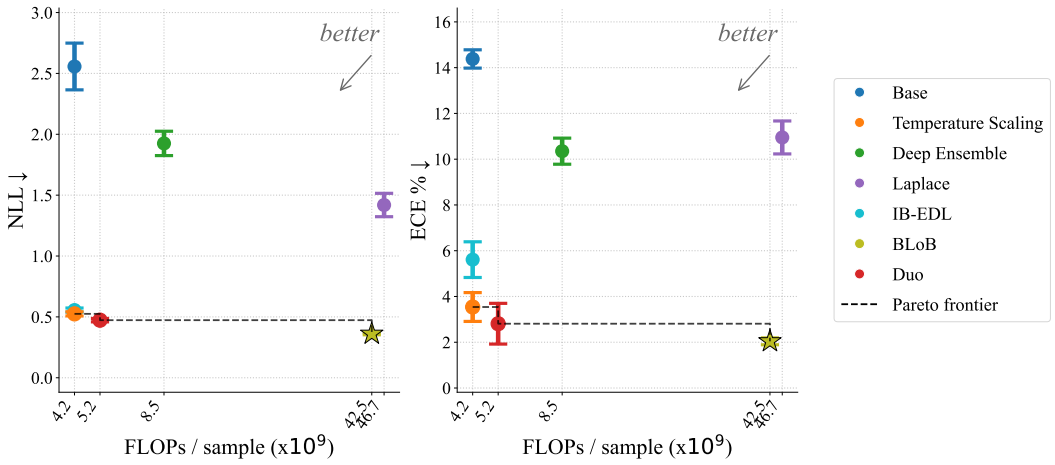


Figure 1: Pareto frontier of floating-point operations (FLOPs) vs. negative log-likelihood (NLL; left) and expected calibration error (ECE; right) for the CSQA dataset. For both metrics, Temperature Scaling, Duo, and BLoB lie on Pareto frontier. The method with the best NLL/ECE performance is marked with a star.

estimation can be improved with only a modest computational overhead relative to the larger base model.

Since Zhou et al. [2025] exclusively focused on computer vision tasks and models, it remains unclear whether and how Asymmetric Duos can work for LLM classifiers. In this paper, we pursue this question for LLMs fine-tuned to multiple-choice question answering (MCQA). Specifically, we examine whether asymmetric fusion improves the calibration of LLMs in MCQA and whether it changes the predictive behavior of the base model in a systematic and cost-efficient way. This positions the paper at the intersection of LLM calibration, uncertainty-aware adaptation, and efficient multi-model uncertainty estimation. Our contributions are organized as follows:

1. **Cost-efficient calibration:** We show that Asymmetric Duos improve LLM calibration on the cost–performance frontier, matching or surpassing state-of-the-art baselines often at a fraction of the inference FLOPs across all benchmarks (cf. Figure 1).
2. **Compatibility with calibration-aware methods:** We show that Asymmetric Duos compose additively with existing calibration-aware approaches such as BLoB, Laplace-LoRA, and IB-EDL, consistently improving calibration performance when applied in composition with these methods.
3. **Logit-space mechanism:** Our mechanistic study sheds light on how asymmetric logit fusion reshapes the logit geometry of the base model, reducing overconfidence by smoothing extreme predictions while preserving decision boundaries, thereby improving calibration without degrading accuracy.

2 Related work & background

Calibration and uncertainty in neural networks and LLMs. The objective of confidence calibration is to match the confidence assigned by a model with the empirical probability of correctness. Although modern neural networks achieve strong predictive performance, this does not necessarily imply well-calibrated confidence estimates, and overconfidence remains a well-known issue [Guo et al., 2017, Jiang et al., 2020, Shorinwa et al., 2024]. This has motivated a large amount of research on uncertainty estimation and post-hoc calibration methods, with temperature scaling being one of the most widely used and computationally simple approaches [Guo et al., 2017]. In addition, evidential modeling approaches aim to capture predictive uncertainty directly within the model rather than adjusting confidence after training [Sensoy et al., 2018].

With the increasing usage of LLMs, the importance of calibration has become more pronounced. While LLMs achieve strong performance across a wide range of tasks, their uncertainty estimates may still be unreliable, particularly after adaptation to downstream tasks [He et al., 2023, Li et al., 2025]. Moreover, uncertainty estimation in LLMs is influenced by additional factors such as prompting, decoding strategies, and answer extraction, making calibration more complex than in classical neural networks [Zhang et al., 2024].

Multiple-choice question answering as a testbed for LLM uncertainty. MCQA is a prominent use case of LLMs for evaluating uncertainty since it reduces free-form generation to a finite decision space. This enables a meaningful comparison of confidence scores across different models and methods [Wang et al., 2025, Kumar et al., 2023]. At the same time, the MCQA evaluation setup is subject to several known factors that can influence model behavior. Early work revealed that model predictions can be influenced by the ordering of answer options, as well as selection bias associated with token usage [Pezeshkpour and Hruschka, 2024, Wei et al., 2024]. Moreover, according to recent research, even the tokenization in next-token response extraction impacts MCQA results and calibration estimates [Sanz-Guerrero et al., 2025]. Thus, with all its advantages and disadvantages, MCQA offers a suitable framework for controlled comparison between different models and methods.

Fine-tuning, parameter-efficient adaptation, and calibration-aware training. One of the main techniques used to adapt LLMs to downstream tasks is fine-tuning. Parameter-efficient techniques such as Low-Rank Adaptation (LoRA) allow for model adaptation at significantly lower computational costs compared to full fine-tuning [Hu et al., 2022, Ding et al., 2023]. However, the roles of adaptation methods and calibration methods are fundamentally different. While parameter-efficient fine-tuning focuses on improving the specialization of a model for a certain task, calibration aims to improve the confidence estimates generated by the model. Previous work has indicated that fine-tuned language models can still remain overconfident, motivating research into calibration-aware training procedures in addition to the typical adaptation process [He et al., 2023]. This has also encouraged the extension of evidential methods to modern LLMs. For instance, Li et al. [2025] proposed IB-EDL as a calibration-aware training method for fine-tuned LLMs.

Multi-model uncertainty estimation and Asymmetric Duos. Ensemble-based approaches improve uncertainty estimation by combining predictions from multiple models, but standard deep ensembles are computationally expensive because they require training and evaluating several models for each prediction [Lakshminarayanan et al., 2017, Fort et al., 2019]. This cost has motivated more efficient uncertainty-aware alternatives. In the LLM setting, recent work has focused on uncertainty estimation through LoRA ensembles and Bayesian parameter-efficient adaptation methods. For example, Laplace-LoRA applies a Laplace approximation to LoRA adapters for post-hoc Bayesian inference [Yang et al., 2024a, Daxberger et al., 2021], while BLoB performs Bayesian LoRA through variational inference during fine-tuning [Wang et al., 2024]. Ensemble-based approaches for LoRA-adapted LLMs have also been explored [Balabanov and Linander, 2025, Wang et al., 2023]. These approaches aim to improve uncertainty estimation while remaining compatible with LoRA-style fine-tuning.

Asymmetric Duos provide a complementary strategy. Zhou et al. [2025] pair a strong base model with a much smaller sidekick model and combine their logits through learned weighting, improving uncertainty estimation with only a small fraction of additional computation in computer vision settings.

3 Asymmetric LLM Duos

Our work builds on the idea of Zhou et al. [2025] by studying whether asymmetric logit fusion transfers to LLM-based MCQA. In addition, we investigate whether the Asymmetric Duos setup can further improve performance when combined with existing calibration-aware methods such as BLoB and Laplace-LoRA.

3.1 Problem formulation

Let $\mathbf{x} \in \mathcal{X}$ denote the input and $y \in \mathcal{Y}$ the corresponding label, where $|\mathcal{Y}| = K$. We consider two predictive models: a base model f_{base} and a sidekick model f_{side}. Each model produces a logit

Algorithm 1 Asymmetric Duo training and inference

Require: $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{test}}$, models $f_{\text{base}}, f_{\text{side}}$

Train $f_{\text{base}}, f_{\text{side}}$ on $\mathcal{D}_{\text{train}}$ ▷ *Independent fine-tuning*

for each $(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{val}}$ **do**

  $\mathbf{z}_{\text{base}}^{(i)} \leftarrow f_{\text{base}}(\mathbf{x}_i)$

  $\mathbf{z}_{\text{side}}^{(i)} \leftarrow f_{\text{side}}(\mathbf{x}_i)$ ▷ *Logit extraction*

end for

$(\alpha^*, \beta^*) = \arg\min_{\alpha,\beta \geq 0} \mathcal{L}(\alpha, \beta)$ ▷ *Weight optimization (Eqs. (1) and (2))*

for each $\mathbf{x} \in \mathcal{D}_{\text{test}}$ **do** ▷ *Inference*

  $\mathbf{z}_{\text{duo}}(\mathbf{x}) \leftarrow \alpha^* \mathbf{z}_{\text{base}}(\mathbf{x}) + \beta^* \mathbf{z}_{\text{side}}(\mathbf{x})$

  $\hat{y} \leftarrow \arg\max_{y \in \mathcal{Y}} \ \text{softmax}(\mathbf{z}_{\text{duo}}(\mathbf{x}))$

end for

vector over the label space, $\mathbf{z}_{\text{base}}(\mathbf{x}) \in \mathbb{R}^K$ and $\mathbf{z}_{\text{side}}(\mathbf{x}) \in \mathbb{R}^K$, which yield class probability vectors $p_{\text{base}}(y \mid \mathbf{x}) = \text{softmax}(\mathbf{z}_{\text{base}}(\mathbf{x}))$ and $p_{\text{side}}(y \mid \mathbf{x}) = \text{softmax}(\mathbf{z}_{\text{side}}(\mathbf{x}))$. Our goal is to construct a combined predictor that improves the quality of uncertainty estimates while maintaining predictive performance.

Following [Zhou et al., 2025], we define the Asymmetric Duo predictor via a linear combination in logit space, $\mathbf{z}_{\text{duo}}(\mathbf{x}) = \alpha\, \mathbf{z}_{\text{base}}(\mathbf{x}) + \beta\, \mathbf{z}_{\text{side}}(\mathbf{x})$. These combined logits parameterize a categorical distribution over answer choices, with class probabilities given by $p_{\text{duo}}(y \mid \mathbf{x}) = \text{softmax}(\mathbf{z}_{\text{duo}}(\mathbf{x}))$. The final prediction is obtained as $\hat{y}(\mathbf{x}) = \arg\max_{y \in \mathcal{Y}} p_{\text{duo}}(y \mid \mathbf{x})$. The weights α and β are global parameters shared across all inputs. We deliberately restrict the fusion to global, input-independent weights in order to avoid introducing additional model complexity and training overhead.

3.2 Learning the fusion weights

Let $\mathcal{D}_{\text{val}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denote the validation set. For each validation example, we precompute and store the aligned logit vectors $\mathbf{z}_{\text{base}}^{(i)}$ and $\mathbf{z}_{\text{side}}^{(i)}$. We then estimate the fusion scales α and β by solving the optimization problem

$$(\alpha^*, \beta^*) = \arg \min_{\alpha,\beta \geq 0} \mathcal{L}(\alpha, \beta), \tag{1}$$

where the validation negative log-likelihood (NLL) is defined as

$$\mathcal{L}(\alpha, \beta) = -\frac{1}{N} \sum_{i=1}^{N} \log \left[\text{softmax} \left(\alpha \mathbf{z}_{\text{base}}^{(i)} + \beta \mathbf{z}_{\text{side}}^{(i)} \right) \right]_{y_i}. \tag{2}$$

In our implementation, this is a two-dimensional post-hoc optimization problem solved with L-BFGS-B, a quasi-Newton method for bound-constrained optimization [Zhu et al., 1997]. The fusion scales are not constrained to sum to one. Instead, we only enforce non-negativity through lower bounds on α and β. Thus, α and β should be interpreted as effective logit scales rather than normalized mixture weights.

The logits of both the base and sidekick models are matched by example index before optimization, ensuring that the two models are combined on the same inputs. Since this procedure uses only saved validation predictions, the Duo can be constructed after independent fine-tuning without any joint training of the two models; Algorithm 1 gives the complete training and inference pipeline.

3.3 Interpretation

The learned weights α and β determine the relative contributions of the base and sidekick models across the dataset. Since the optimization is driven by validation NLL, the resulting combination reflects which model provides more informative or better-calibrated logits. If the sidekick model provides little useful information, the optimization assigns it a small weight, effectively recovering the base model. Conversely, when the sidekick model captures complementary patterns, its contribution increases and improves the Duo predictor. In our experiments, we empirically validate this behavior in Table 1, where we observe consistent improvements in NLL and expected calibration error (ECE) across all datasets.

Implicit per-model temperature scaling. Beyond inter-model weighting, the unconstrained non-negative parameterization of (α, β) has a direct calibration interpretation: each scale acts as an inverse temperature on its own model's logits. Concretely, $\alpha\,\mathbf{z}_{\text{base}}(\mathbf{x}) = \mathbf{z}_{\text{base}}(\mathbf{x})/T_{\text{base}}$ with $T_{\text{base}} = 1/\alpha$, and analogously for the sidekick model. The validation-NLL minimization in Eq. (1) therefore performs two operations in a single two-dimensional fit: (i) per-model temperature scaling [Guo et al., 2017], and (ii) inter-model relative weighting. This is in contrast with formulations that fix $\alpha + \beta = 1$, which pin the joint logit magnitude and decouple it from any temperature effect. By keeping (α, β) unconstrained on the non-negative orthant, Duo includes both calibration steps without an extra optimization stage. Consequently, a Duo built from temperature-scaled base and sidekick logits, $\mathbf{z}_{\text{duo}}(\mathbf{x}) = \alpha\,\mathbf{z}_{\text{base}}(\mathbf{x})/T_{\text{base}} + \beta\,\mathbf{z}_{\text{side}}(\mathbf{x})/T_{\text{side}}$, can always be reparameterized as $\tilde{\alpha}\,\mathbf{z}_{\text{base}}(\mathbf{x}) + \tilde{\beta}\,\mathbf{z}_{\text{side}}(\mathbf{x})$ with $\tilde{\alpha} = \alpha/T_{\text{base}}$ and $\tilde{\beta} = \beta/T_{\text{side}}$, effectively recovering the same predictor as the plain Duo.

4 Experiments

We now study our adaptation of the Asymmetric Duo for LLM MCQA.

4.1 Experimental setups

Models. We fine-tune two Qwen2 [Yang et al., 2024b] family models with parameters 7B and 1.5B using LoRA [Hu et al., 2022] implemented using PEFT [Mangrulkar et al., 2022] and Transformers [Wolf et al., 2020]. Implementation details and hyperparameter choices can be found in Appendix C. After this section, these two models will be referred to as the base and sidekick models and abbreviated as B and S, respectively.

Datasets. In the experiments, six MCQA datasets are used, which are ARC-C and ARC-E [Clark et al., 2018], OpenBookQA (OBQA) [Mihaylov et al., 2018], CommonsenseQA (CSQA) [Talmor et al., 2019], SciQ [Welbl et al., 2017] and RACE [Lai et al., 2017]. These datasets are chosen to align with experimental settings in prior work, facilitating consistent and comparable evaluation across different methods [Wang et al., 2024, Hu et al., 2022, Li et al., 2025]. The target space is defined as tokens corresponding to different pre-determined options like A, B, C, D, and E.

Baselines. Our method is compared to six different baselines: LoRA fine-tuned MAP predictions (**B**), a deep ensemble with two members (**B DE**) [Lakshminarayanan et al., 2017, Fort et al., 2019], Temperature Scaling (**B TS**) [Guo et al., 2017], Laplace-LoRA (**B LA**) [Yang et al., 2024a], IB-EDL (**B IB-EDL**) [Li et al., 2025], and BLoB (**B BLoB**) [Wang et al., 2024]. In order to test whether applying the Duo method to existing uncertainty-aware approaches can further lead to a performance boost, we additionally applied the Duo procedure to the outputs of the following baselines: Laplace-LoRA (**Duo LA**), IB-EDL (**Duo IB-EDL**), and BLoB (**Duo BLoB**).

Measuring the computational cost. We use FLOPs to measure inference cost, as they provide a hardware-independent metric for comparing computational complexity across different methods. FLOPs are estimated from multiply-accumulate operations (MACs), where one MAC corresponds to two FLOPs. For the Duo framework, FLOPs are computed as the sum of the base and sidekick model inference costs. This allows for a fair comparison against baselines such as single models and deep ensembles of varying sizes. All estimates are obtained from standard model-level complexity calculations based on the model architecture and input sequence length.

4.2 Cost-efficient uncertainty quantification

An effective uncertainty-aware method should both enhance model calibration and maintain predictive accuracy comparable to standard MAP training [Li et al., 2025]. Therefore, we use accuracy (Acc), ECE, and NLL as evaluation metrics on the six aforementioned datasets.

As shown in Table 1, Duo is highly competitive across all metrics and datasets compared to baselines. Duo achieves the best NLL on the ARC-C dataset, providing an almost 78% decrease on average compared to Base ($2.075 \rightarrow 0.469$). For the other five datasets, Duo achieves the second best performance, with only small gaps relative to the best performing method; for example, on ARC-E,

Table 1: Calibration performance of baseline and Duo methods fine-tuned on Qwen2 7B ("B" for Base) and Qwen2 1.5B ("S" for Sidekick). Arrows (↑ or ↓) indicate the direction of better performance for each metric. The best results are highlighted in **bold**, and the second-best results are underlined. Additionally, ± denotes the standard deviation over 5 runs with 5 different seeds. The Duo method has either the best or the second-best performance across most of the datasets and metrics.

Metric	Method	ARC-C	ARC-E	CSQA	OBQA	RACE	SciQ
NLL ↓	S	$4.234_{\pm 0.046}$	$1.968_{\pm 0.100}$	$3.493_{\pm 0.182}$	$1.497_{\pm 0.076}$	$0.590_{\pm 0.021}$	$0.546_{\pm 0.022}$
	B	$2.075_{\pm 0.041}$	$0.655_{\pm 0.065}$	$2.557_{\pm 0.192}$	$0.991_{\pm 0.080}$	$0.340_{\pm 0.009}$	$0.333_{\pm 0.018}$
	B TS	$0.496_{\pm 0.011}$	$0.213_{\pm 0.015}$	$0.525_{\pm 0.017}$	$0.357_{\pm 0.012}$	$0.268_{\pm 0.005}$	$0.183_{\pm 0.008}$
	B DE	$1.682_{\pm 0.069}$	$0.538_{\pm 0.029}$	$1.925_{\pm 0.100}$	$0.750_{\pm 0.054}$	$0.319_{\pm 0.006}$	$0.295_{\pm 0.007}$
	B LA	$1.212_{\pm 0.052}$	$0.401_{\pm 0.029}$	$1.419_{\pm 0.096}$	$0.602_{\pm 0.068}$	$0.308_{\pm 0.006}$	$0.259_{\pm 0.016}$
	B IB-EDL	$0.547_{\pm 0.018}$	$0.220_{\pm 0.021}$	$0.556_{\pm 0.017}$	$0.368_{\pm 0.016}$	$0.267_{\pm 0.003}$	$0.205_{\pm 0.005}$
	B BLoB	$1.347_{\pm 0.101}$	$\mathbf{0.183_{\pm 0.039}}$	$\mathbf{0.361_{\pm 0.008}}$	$\mathbf{0.240_{\pm 0.016}}$	$\mathbf{0.247_{\pm 0.009}}$	$\mathbf{0.136_{\pm 0.002}}$
	Duo	$\mathbf{0.469_{\pm 0.009}}$	$0.188_{\pm 0.014}$	$0.473_{\pm 0.014}$	$0.302_{\pm 0.007}$	$0.262_{\pm 0.006}$	$0.167_{\pm 0.006}$
Acc ↑	S	$71.54_{\pm 0.90}$	$85.51_{\pm 0.61}$	$74.63_{\pm 0.27}$	$80.32_{\pm 0.52}$	$84.29_{\pm 0.42}$	$90.04_{\pm 0.15}$
	B	$86.59_{\pm 0.52}$	$95.08_{\pm 0.40}$	$85.34_{\pm 0.41}$	$90.24_{\pm 0.17}$	$91.89_{\pm 0.11}$	$94.96_{\pm 0.21}$
	B TS	$86.59_{\pm 0.52}$	$95.08_{\pm 0.40}$	$85.34_{\pm 0.41}$	$90.24_{\pm 0.17}$	$91.89_{\pm 0.11}$	$94.96_{\pm 0.21}$
	B DE	$\mathbf{87.15_{\pm 0.43}}$	$95.09_{\pm 0.14}$	$85.81_{\pm 0.35}$	$90.20_{\pm 0.31}$	$\mathbf{91.94_{\pm 0.23}}$	$95.12_{\pm 0.25}$
	B LA	$86.66_{\pm 0.48}$	$95.11_{\pm 0.32}$	$85.32_{\pm 0.42}$	$90.04_{\pm 0.14}$	$91.80_{\pm 0.15}$	$94.84_{\pm 0.23}$
	B IB-EDL	$86.28_{\pm 0.60}$	$\mathbf{95.23_{\pm 0.17}}$	$85.39_{\pm 0.52}$	$90.84_{\pm 0.38}$	$91.85_{\pm 0.26}$	$95.02_{\pm 0.22}$
	B BLoB	$86.31_{\pm 0.67}$	$94.89_{\pm 0.38}$	$\mathbf{86.55_{\pm 0.61}}$	$\mathbf{91.84_{\pm 0.83}}$	$91.57_{\pm 0.59}$	$\mathbf{95.42_{\pm 0.08}}$
	Duo	$86.81_{\pm 0.49}$	$95.14_{\pm 0.40}$	$85.73_{\pm 0.42}$	$90.60_{\pm 0.14}$	$91.82_{\pm 0.14}$	$95.24_{\pm 0.26}$
ECE ↓	S	$27.73_{\pm 0.96}$	$14.16_{\pm 0.59}$	$24.61_{\pm 0.36}$	$18.57_{\pm 0.65}$	$10.11_{\pm 0.47}$	$8.69_{\pm 0.14}$
	B	$13.05_{\pm 0.52}$	$4.74_{\pm 0.42}$	$14.38_{\pm 0.40}$	$9.55_{\pm 0.32}$	$5.63_{\pm 0.15}$	$4.75_{\pm 0.13}$
	B TS	$4.35_{\pm 0.84}$	$1.53_{\pm 0.21}$	$3.54_{\pm 0.63}$	$4.13_{\pm 1.07}$	$2.73_{\pm 0.30}$	$3.07_{\pm 0.53}$
	B DE	$10.11_{\pm 0.93}$	$3.76_{\pm 0.13}$	$10.35_{\pm 0.57}$	$7.14_{\pm 0.22}$	$5.17_{\pm 0.25}$	$4.12_{\pm 0.07}$
	B LA	$9.79_{\pm 0.39}$	$3.55_{\pm 0.31}$	$10.95_{\pm 0.72}$	$7.03_{\pm 0.42}$	$4.77_{\pm 0.23}$	$3.81_{\pm 0.25}$
	B IB-EDL	$4.88_{\pm 1.04}$	$3.90_{\pm 1.01}$	$5.61_{\pm 0.78}$	$9.37_{\pm 1.20}$	$3.84_{\pm 0.55}$	$2.89_{\pm 0.65}$
	B BLoB	$12.27_{\pm 0.67}$	$5.48_{\pm 2.90}$	$\mathbf{2.04_{\pm 0.15}}$	$\mathbf{2.71_{\pm 0.75}}$	$\mathbf{2.17_{\pm 0.31}}$	$\mathbf{1.55_{\pm 0.38}}$
	Duo	$\mathbf{3.90_{\pm 1.44}}$	$\mathbf{1.30_{\pm 0.15}}$	$2.81_{\pm 0.89}$	$3.09_{\pm 0.53}$	$2.78_{\pm 0.37}$	$1.94_{\pm 0.42}$

B BLoB obtains an average NLL of 0.183, while Duo achieves 0.188 and their confidence intervals overlap. Calibration gains also hold for ECE. Duo provides a reduction of approximately $60 - 80\%$ compared to Base. Additionally, for five of the six datasets, Duo is either the best or second-best-performing method, with RACE being the only dataset on which it does not achieve this ranking. However, its standard deviation band overlaps with that of the second-best-performing method, which is B TS. Despite consistently improving the calibration, Duo also preserves the predictive accuracy of Base. Across all datasets, Duo either improves accuracy or maintains it at a comparable level. From these perspectives, it can be considered as an effective method of predicting uncertainty.

FLOP-Performance tradeoff. The main competitiveness of Duo comes from its computational efficiency. Unlike other uncertainty-aware methods that require costly ensembles, repeated stochastic forward passes, or additional Bayesian approximation procedures, Duo achieves similar, and sometimes better, calibration improvements with substantially lower computational overhead. At the same time, it preserves the predictive accuracy of the Base model, making it a practical and scalable alternative for an uncertainty-aware calibration method. As illustrated in Figure 1, the Pareto frontier highlights the trade-off between performance and computational cost for the CSQA dataset. While B BLoB achieves the best results across metrics, Duo remains highly competitive and lies on the Pareto frontier with only one-eighth of the computational cost. This is particularly evident when comparing FLOPs per sample, where Duo offers a favorable balance between ECE, and NLL. This also holds for other datasets provided in Appendix A.1.

4.3 Compatibility with calibration-aware methods

To better understand the extent of performance improvements of the Duo, we next ask whether the Duo procedure composes additively with calibration-aware baseline methods. For each calibration family $X \in \{\text{LA}, \text{BLoB}, \text{IB-EDL}\}$ we form a Duo whose base and sidekick models are both calibrated with X and learn (α, β) on the validation split, at a marginal cost of $\approx 1.22\times$ the inference FLOPs of B X. Figure 2 reports the relative NLL and ECE reduction of Duo X over B X. Full numerical results, including standard deviations and accuracy, are deferred to Table 4 in Appendix B.1. We also exclude

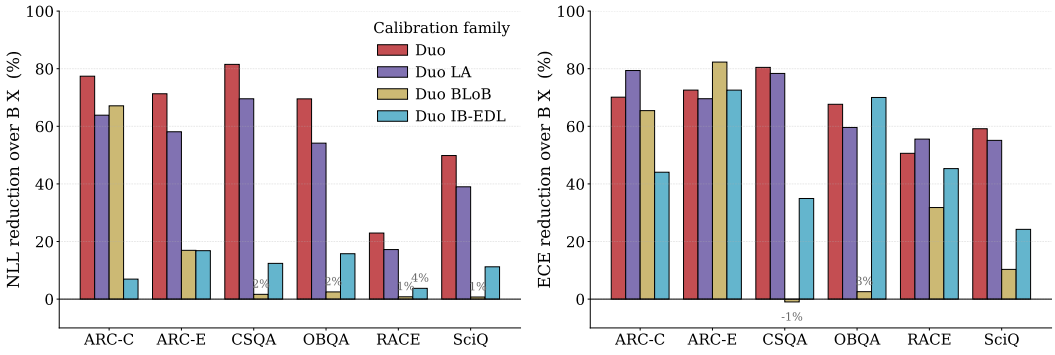


Figure 2: Relative reduction (%) of NLL (left) and ECE (right) achieved by Duo X over its base predictor B X across the six MCQA benchmarks.

a Duo formed from temperature-scaled base and sidekick models, as it reduces to the standard Duo by reparameterization (cf. Section 3.3). The three families behave differently, and the improvements separate into two regimes set by how well-calibrated the underlying base already is.

In the weakly calibrated regime, Laplace-LoRA only mildly recalibrates the overconfident Base, so B LA remains overconfident. Duo LA over B LA produces large reductions in NLL and ECE with non-overlapping SD bands in Table 4, mirroring the Duo-over-Base improvements quantified in Section 4.2. In both cases, the optimizer converges to $\alpha, \beta \ll 1$ (Table 5), so Duo is performing global temperature scaling in addition to combining the two predictors.

The strongly calibrated regime diverges from this pattern. BLoB and IB-EDL already calibrate the base model substantially before any fusion, in several cells matching or beating plain Duo, so the room left for Duo to improve is smaller and many cells of Table 4 now have margins within one SD. The optimizer does not compress these base models; it converges to $\alpha \approx 1$ for both families and instead uses β to decide how much to incorporate the sidekick. For Duo BLoB, β collapses to (near) zero on RACE and SciQ, so Duo BLoB reduces to B BLoB on those datasets, visible in Figure 2 as the bars closest to zero. For Duo IB-EDL, β stays above zero, so the sidekick still contributes, and the relative gains remain consistent but smaller than those for LA.

4.4 Ablation: Trivial Duos

The Duo optimization stage is essential. Table 2 replaces the learned (α, β) with two trivial settings, $\alpha = \beta = 0.5$ (logit averaging) and $\alpha = \beta = 1$ (logit sum). Unlike the optimized Duo formulation, these variants rely on fixed fusion coefficients and therefore cannot benefit from either global temperature scaling or asymmetric model weighting.

As a result, NLL and ECE degrade markedly across all benchmarks. Compared to the optimized Duo, trivial averaging increases NLL by roughly 2–3$\times$ and ECE by approximately 2–5$\times$ across datasets, while the unscaled logit sum leads to even larger degradations. The two trivial settings produce identical accuracy because softmax is invariant under uniform logit scaling, but their fixed weights cannot encode the asymmetric reliability of B and S; only the validation-tuned weights downweight the weaker Sidekick.

5 Mechanistic logit-space analysis of Asymmetric Duos

In this section, we analyze how Duo improves the calibration of LLMs by examining changes in their logit-level behavior. In particular, we compare the logit distributions of the base and sidekick models and Duo. For clarity, we focus on the CSQA dataset using a single seed, while results for the remaining datasets are provided in Figures 10–19 in Appendix A.2 and Table 6 in Appendix B.3.

Let $\mathcal{D}_{\text{test}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ denote the test set, where $\mathbf{x}_i$ is the input and $y_i \in \{1, \ldots, K\}$ is the correct class. For each input $\mathbf{x}_i$, the model outputs logits $\mathbf{z}_i = (z_{i,1}, z_{i,2}, \ldots, z_{i,K}) \in \mathbb{R}^K$, where $z_{i,k}$

Table 2: Ablation on the Duo fusion weights. Trivial Duos (Duo (0.5)) with $\alpha = \beta = 0.5$ (logit averaging) and $\alpha = \beta = 1$ (Duo (1); logit sum) skip the Duo optimization stage. The best method per metric–dataset combination is marked in **bold**. All metrics are mean ± SD over 5 repetitions.

Metric	Method	ARC-C	ARC-E	CSQA	OBQA	RACE	SciQ
NLL ↓	Duo (0.5)	$1.442_{\pm 0.052}$	$0.459_{\pm 0.042}$	$1.489_{\pm 0.103}$	$0.523_{\pm 0.029}$	$0.315_{\pm 0.008}$	$0.236_{\pm 0.012}$
	Duo (1)	$2.437_{\pm 0.072}$	$0.805_{\pm 0.070}$	$2.481_{\pm 0.105}$	$0.977_{\pm 0.059}$	$0.531_{\pm 0.015}$	$0.437_{\pm 0.025}$
	Duo	$\mathbf{0.469}_{\pm 0.009}$	$\mathbf{0.188}_{\pm 0.014}$	$\mathbf{0.473}_{\pm 0.014}$	$\mathbf{0.302}_{\pm 0.007}$	$\mathbf{0.262}_{\pm 0.006}$	$\mathbf{0.167}_{\pm 0.006}$
Acc ↑	Duo (0.5)	$84.32_{\pm 0.84}$	$93.48_{\pm 0.48}$	$85.11_{\pm 0.54}$	$90.36_{\pm 0.36}$	$91.03_{\pm 0.17}$	$95.14_{\pm 0.27}$
	Duo (1)	$84.32_{\pm 0.84}$	$93.48_{\pm 0.48}$	$85.11_{\pm 0.54}$	$90.36_{\pm 0.36}$	$91.03_{\pm 0.17}$	$95.14_{\pm 0.27}$
	Duo	$\mathbf{86.81}_{\pm 0.49}$	$\mathbf{95.14}_{\pm 0.40}$	$\mathbf{85.73}_{\pm 0.42}$	$\mathbf{90.60}_{\pm 0.14}$	$\mathbf{91.82}_{\pm 0.14}$	$\mathbf{95.24}_{\pm 0.26}$
ECE ↓	Duo (0.5)	$11.64_{\pm 0.67}$	$4.30_{\pm 0.36}$	$12.63_{\pm 0.36}$	$6.71_{\pm 0.27}$	$4.46_{\pm 0.21}$	$3.43_{\pm 0.43}$
	Duo (1)	$13.56_{\pm 0.76}$	$5.24_{\pm 0.33}$	$13.84_{\pm 0.36}$	$8.04_{\pm 0.29}$	$6.41_{\pm 0.24}$	$4.34_{\pm 0.40}$
	Duo	$\mathbf{3.90}_{\pm 1.44}$	$\mathbf{1.30}_{\pm 0.15}$	$\mathbf{2.81}_{\pm 0.89}$	$\mathbf{3.09}_{\pm 0.53}$	$\mathbf{2.78}_{\pm 0.37}$	$\mathbf{1.94}_{\pm 0.42}$

denotes the logit assigned to class k. The predicted class is $\hat{y}_i = \arg\max_{k \in \{1,\ldots,K\}} p_{i,k}$, and the corresponding predictive probabilities are obtained using the softmax function $p_{i,k} = \text{softmax}(\mathbf{z}_i)$.

We consider the following three quantities in our analysis:

- The logit margin, $z_{i,y_i} - \max_{k \neq y_i} z_{i,k}$, which measures the gap between the correct-class logit and the highest incorrect-class logit. Positive values indicate that the correct class receives the highest logit, and negative values indicate that at least one incorrect class receives a higher logit;
- confidence, $p_{i,\hat{y}_i} = \max_k p_{i,k}$, which is the probability assigned to the predicted class;
- the correct-class probability, p_{i,y_i}, which is the probability assigned to the correct class.

5.1 Duo softens Base's overconfidence by compressing logit margins

As can be seen from the empirical confidence CDF in Figure 3, the base model puts nearly all the probability mass into a single class after fine-tuning. This indicates that the base model exhibits highly overconfident behavior, which is also evidenced by poor ECE and NLL metrics, despite maintaining strong predictive performance, as can be seen in Table 1. Figure 3 illustrates how Duo mitigates this overconfidence at the logit level. The left plot shows that the Duo combination compresses the logit margin toward zero, with the median dropping from 22.1 to 3.8. This behavior follows from the unconstrained fusion mechanism: since the weights α and β are optimized without enforcing a simplex constraint, the resulting logits can be rescaled relative to the original models, leading to a softer predictive distribution. The right plot supports this interpretation, where Duo yields smoother, more spread-out probability and confidence distributions, consistent with a model that hedges appropriately rather than concentrating nearly all probability mass on a single answer.

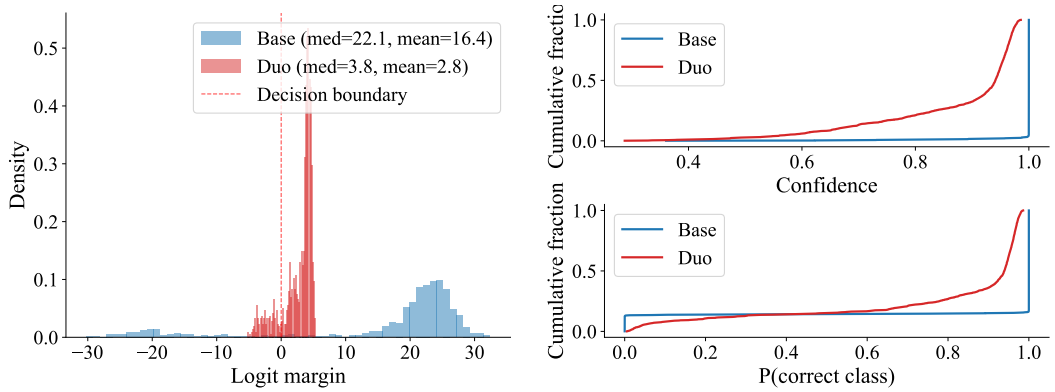


Figure 3: Logit margin (left) and empirical cumulative distribution functions (CDFs) of probability and confidence (right) for Duo and Base on the CSQA dataset.

5.2 Calibration gains come from the wrong predictions of Base

Table 3 decomposes NLL and ECE into samples that Base classifies correctly (Base-Correct) and incorrectly (Base-Wrong). A clear asymmetry can be observed. Pulling the logit margin toward zero lowers p_{i,y_i} on samples that Base already classifies correctly, slightly increasing their per-sample NLL ($0.003 \rightarrow 0.121$). The major improvement comes from misclassified samples: under Base, the strongest competing logit dominates by a wide margin, resulting in a significantly small p_{i,y_i} and producing a large NLL contribution (17.705) that can outweigh correct samples. Duo's compressed margins increase p_{i,y_i} on these samples, collapsing the per-sample NLL to 2.552, roughly a seven-fold reduction. At the same time, Duo recovers the argmax of the Base in most cases, so accuracy (Table 1) remains largely unchanged or slightly improves. Because NLL is unbounded above, the per-sample improvement on incorrect predictions far exceeds the per-sample degradation on correct ones, and the mean drops sharply. The same Base-Correct/Base-Wrong asymmetry appears in ECE ($0.28 \rightarrow 9.18$ versus $98.46 \rightarrow 65.59$), and the mechanism is the same redistribution of probability mass: Base concentrates nearly all of its mass in the top-confidence bin, where the gap between $p_{i,\hat{y}_i}$ and empirical bin accuracy is widest. Duo spreads this mass into intermediate-confidence bins whose accuracy more closely matches their stated confidence, shrinking the calibration gap precisely where it dominates the metric.

Table 3: NLL and ECE of Base and Duo, computed on the CSQA test samples that Base classifies correctly/incorrectly.

Metric	Method	Base Correct	Base Wrong
NLL ↓	Base	**0.003**	17.705
	Duo	0.121	**2.552**
ECE ↓	Base	**0.28**	98.46
	Duo	9.18	**65.59**

6 Discussion

In this study, we propose and study an adaptation of the Asymmetric Duos framework for uncertainty estimation in LLM classifiers as a simple yet cost-efficient alternative to existing methods. Across six MCQA benchmarks, the Duo framework consistently achieves the best or second-best calibration performance, ranking among the top methods in uncertainty quantification metrics such as ECE and NLL, while maintaining or slightly improving the base model's accuracy. Moreover, the Asymmetric Duo framework achieves these improvements at a fraction of the computational cost of stronger baselines such as BLoB, which often achieves better results but requires substantially higher compute. In contrast, the Duo method lies close to the Pareto frontier, offering a favorable trade-off between calibration quality and efficiency.

Beyond standalone performance, we also show that the Duo framework composes additively with other calibration-aware methods. It provides consistent improvements when applied on top of approaches such as Laplace-LoRA, IB-EDL, and BLoB. This suggests that asymmetric fusion of two distinct models captures complementary information that existing single-model techniques do not exploit.

Overall, these results position Asymmetric Duos as a simple yet very effective and broadly applicable approach to improving uncertainty estimation in LLM classifiers. By combining strong performance, low computational overhead, and compatibility with existing calibration techniques, the method offers a practical alternative to more complex and resource-intensive approaches.

Limitations and future work. Our results demonstrate the effectiveness of the Asymmetric Duo framework; however, there remain several limitations. First, we focused on MCQA as a controlled evaluation setting. Extending the analysis to other natural language tasks, such as open-ended generation or sentiment classification, would offer further insights into the behavior of the method in other popular scenarios. This, however, comes with additional challenges such as defining calibration for open-ended sequences. Second, we do not evaluate the framework on current state-of-the-art large-scale models. Investigating performance at larger scales would help clarify the scalability of the approach and its behavior in high-capacity settings such as state-of-the-art commercial LLMs, but requires extensive computational resources and access to closed-source models. Finally, while effective, Duo's requirement for a separate, smaller sidekick model still results in additional memory overhead. A promising direction for future research is to unify the models by employing early exits within the base model's layers to serve the function of the sidekick.

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A Additional figures

A.1 Pareto frontiers

The per-dataset Pareto frontiers in Figures 4–8 confirm that the cost-calibration story discussed for CSQA in the main text is not dataset-specific. Across all six benchmarks, Duo sits in the lower-left "elbow" alongside B TS, achieving comparable NLL and ECE at $\approx 1.22\times$ the FLOPs of the base model and at $\approx 1/8$ the FLOPs of B BLoB. The one partial exception is RACE, where B BLoB and B TS edge Duo on ECE by a small margin (within Duo's standard-deviation band), so Duo is no longer strictly Pareto-dominant there but still lies on the low-cost segment of the frontier.

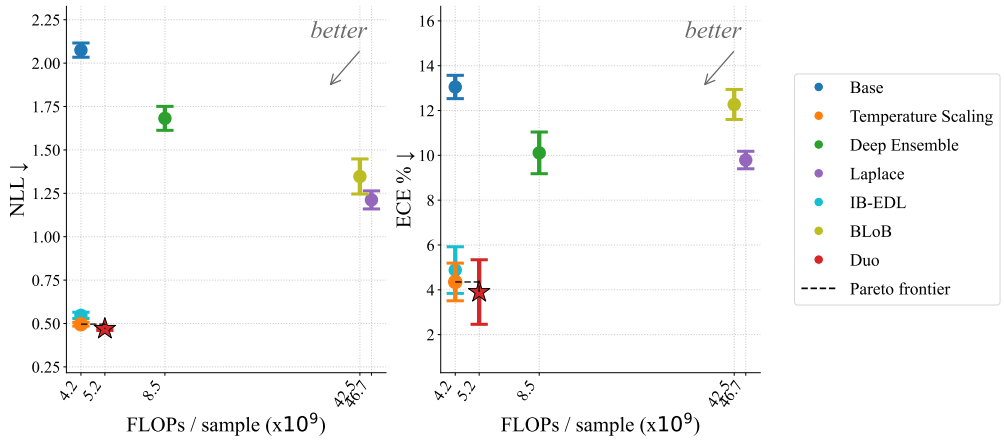


Figure 4: Pareto frontier of FLOPs vs. NLL (left) and ECE (right) for ARC-C dataset. For both metrics, Temperature Scaling and Duo lie on Pareto frontier. The method with the best NLL/ECE performance is marked with a star.

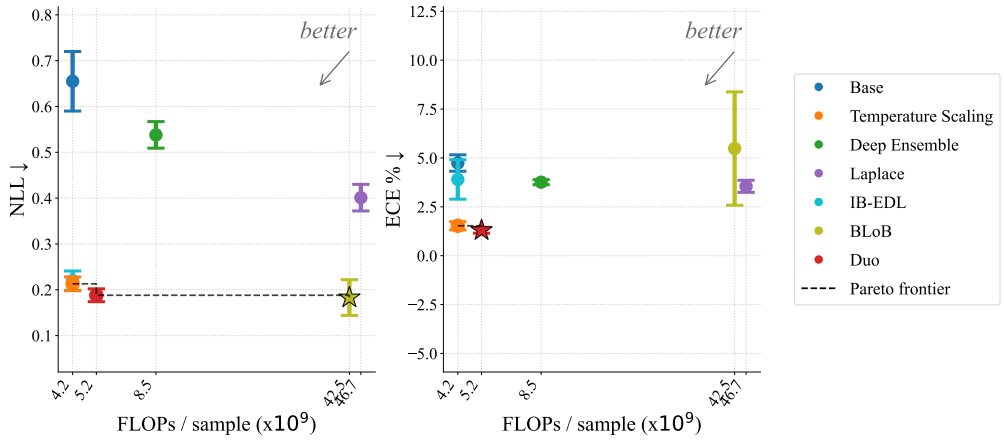


Figure 5: Pareto frontier of FLOPs vs. NLL (left) and ECE (right) for the ARC-E dataset. For NLL, Temperature Scaling, Duo, and BLoB lie on the Pareto frontier; for ECE, Temperature Scaling and Duo lie on the Pareto frontier. The method with the best NLL/ECE performance is marked with a star.

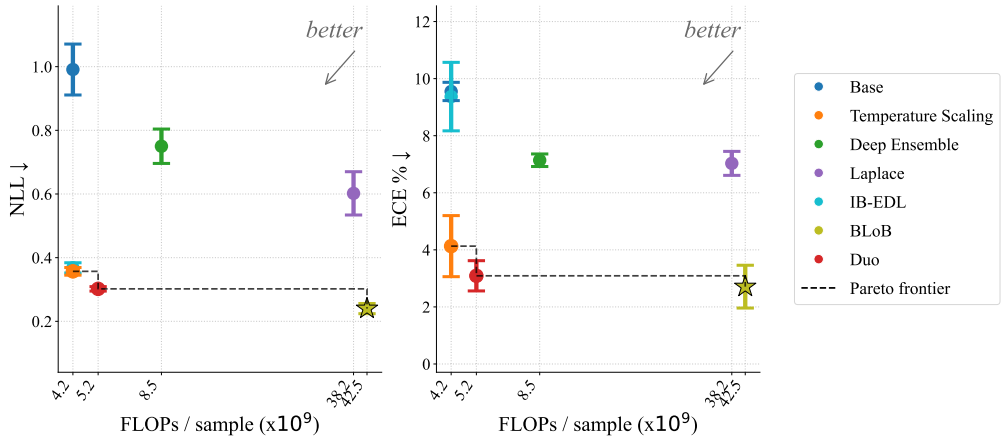


Figure 6: Pareto frontier of FLOPs vs. NLL (left) and ECE (right) for OBQA dataset. For both metrics, Temperature Scaling, Duo, and BLoB lie on Pareto frontier. The method with the best NLL/ECE performance is marked with a star.

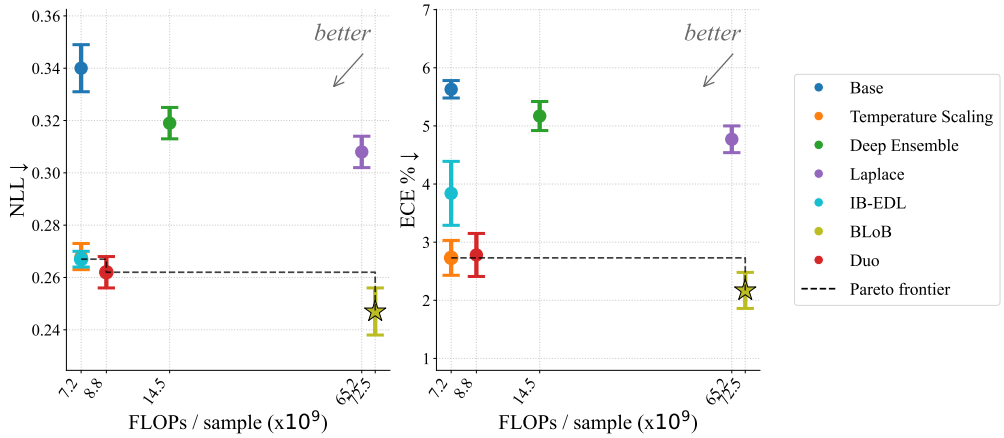


Figure 7: Pareto frontier of FLOPs vs. NLL (left) and ECE (right) for RACE dataset. For NLL, IB-EDL, Duo, and BLoB lie on the Pareto frontier; for ECE, Temperature Scaling and Duo lie on the Pareto frontier. The method with the best NLL/ECE performance is marked with a star.

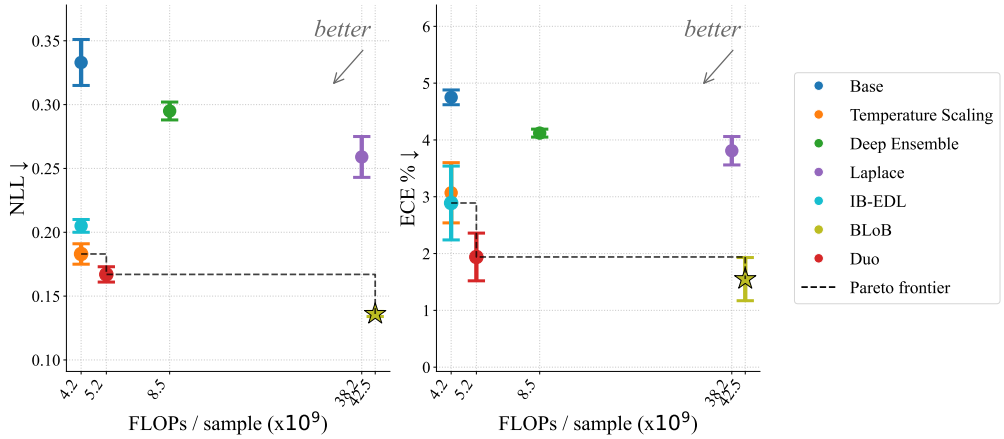


Figure 8: Pareto frontier of FLOPs vs. NLL (left) and ECE (right) for SciQ dataset. For NLL, Temperature Scaling, Duo, and BLoB lie on the Pareto frontier; for ECE, IB-EDL, Duo, and BLoB lie on the Pareto frontier. The method with the best NLL/ECE performance is marked with a star.

A.2 Logit distributions

The per-dataset logit/spread/margin/CDF plots in Figures 10–19 show the same logit-space mechanism we identified for CSQA in Section 5 on every other benchmark. Base and sidekick models concentrate correct-class logits at large positive values with a wide spread, and the Duo predictor systematically pulls the logit margin toward zero. We read this as evidence that asymmetric logit fusion mitigates overconfidence through a single, dataset-agnostic mechanism. The considered quantities are the correct-class logit, z_{i,y_i}, which denotes the logit assigned to the correct class; the logit spread, $\max_k z_{i,k} - \min_k z_{i,k}$, which measures the range of the output logits; as well as the logit margin, confidence, and correct-class probability described in Section 5.

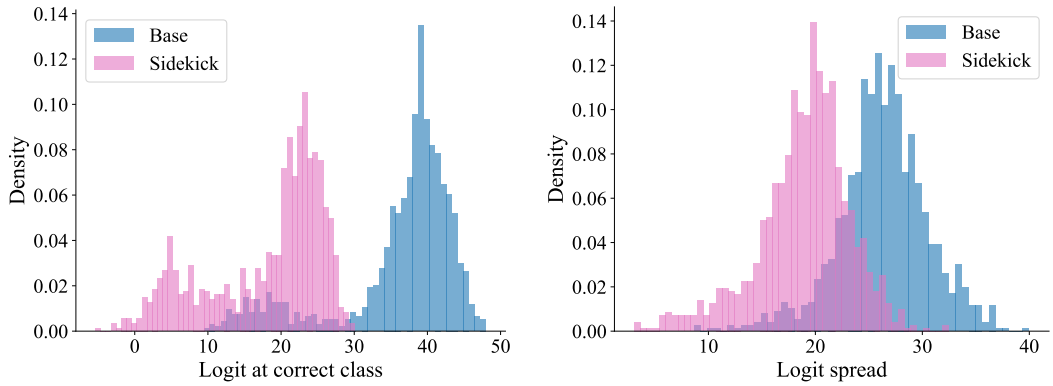


Figure 9: Logit (left) and logit spread (right) distributions of base and sidekick models on the CSQA dataset.

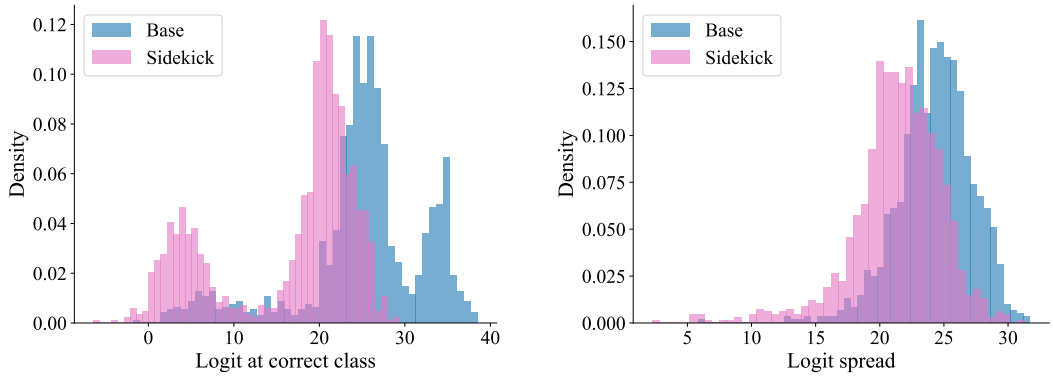


Figure 10: Logit (left) and logit spread (right) distributions of base and sidekick models for ARC-C dataset.

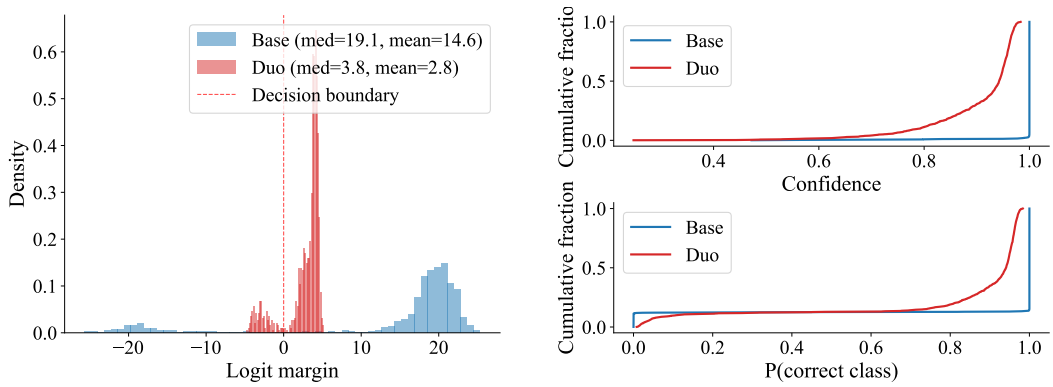


Figure 11: Logit margin (left) and empirical cumulative distribution functions (CDF) of probability and confidence (right) for Duo and Base for ARC-C dataset.

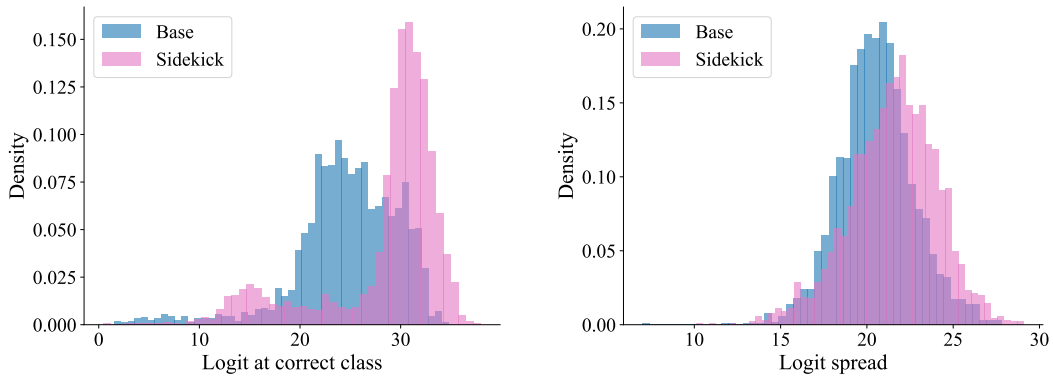


Figure 12: Logit (left) and logit spread (right) distributions of base and sidekick models for ARC-E dataset.

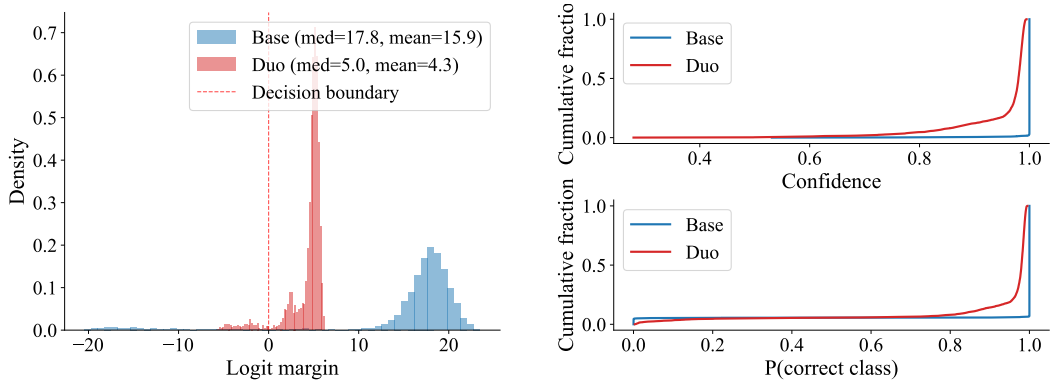


Figure 13: Logit margin (left) and empirical cumulative distribution functions (CDF) of probability and confidence (right) for Duo and Base for ARC-E dataset.

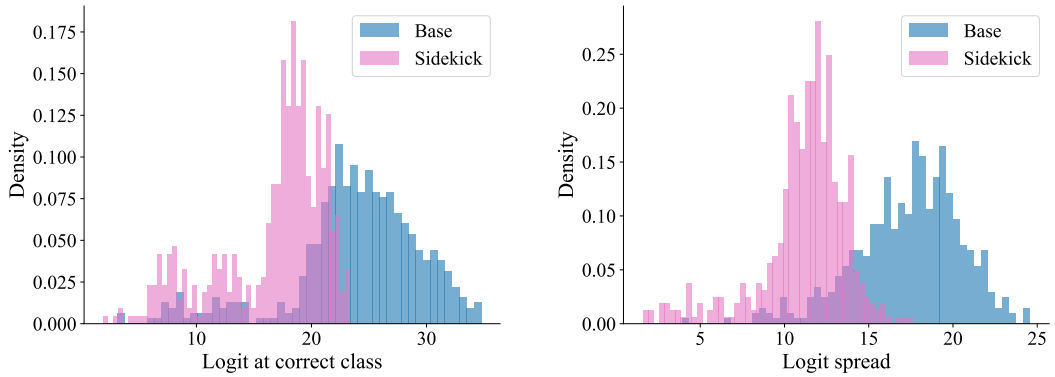


Figure 14: Logit (left) and logit spread (right) distributions of base and sidekick models for OBQA dataset.

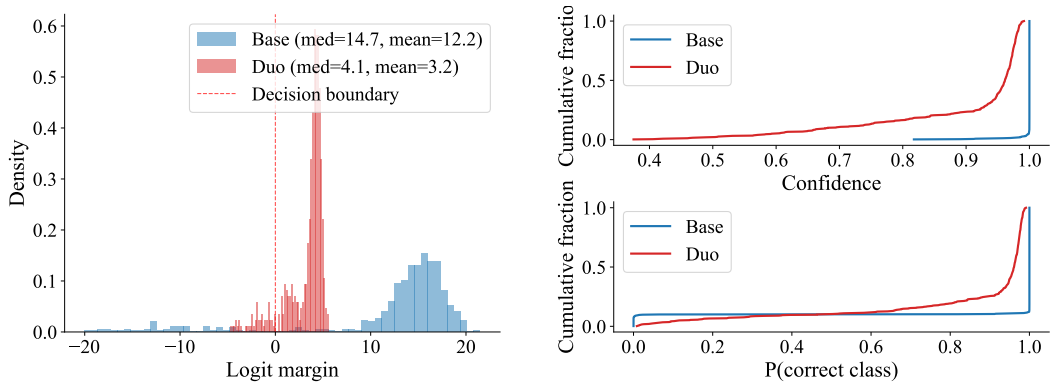


Figure 15: Logit margin (left) and empirical cumulative distribution functions (CDF) of probability and confidence (right) for Duo and Base for OBQA dataset.

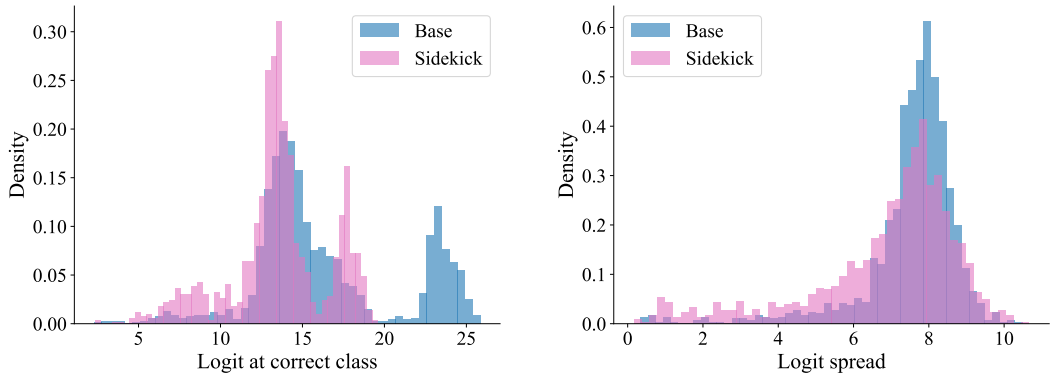


Figure 16: Logit (left) and logit spread (right) distributions of base and sidekick models for RACE dataset.

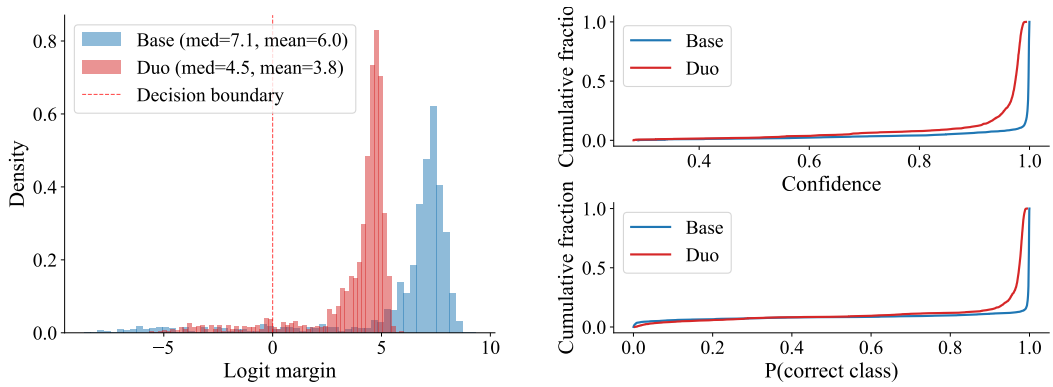


Figure 17: Logit margin (left) and empirical cumulative distribution functions (CDF) of probability and confidence (right) for Duo and Base for RACE dataset.

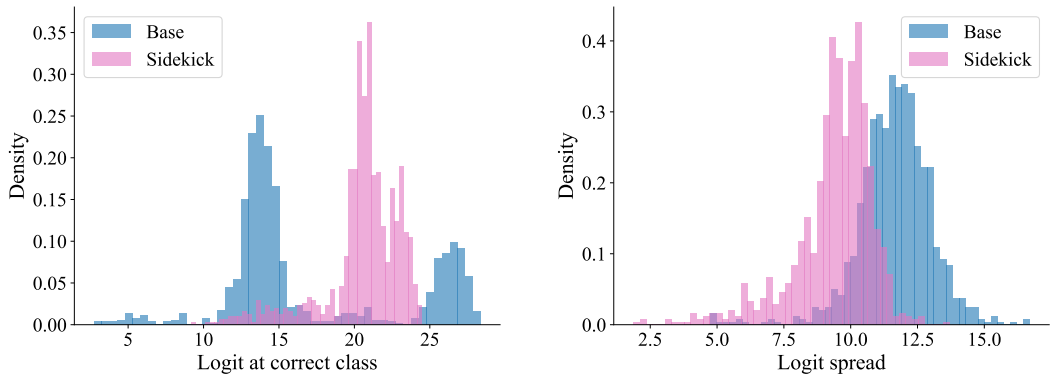


Figure 18: Logit (left) and logit spread (right) distributions of base and sidekick models for SciQ dataset.

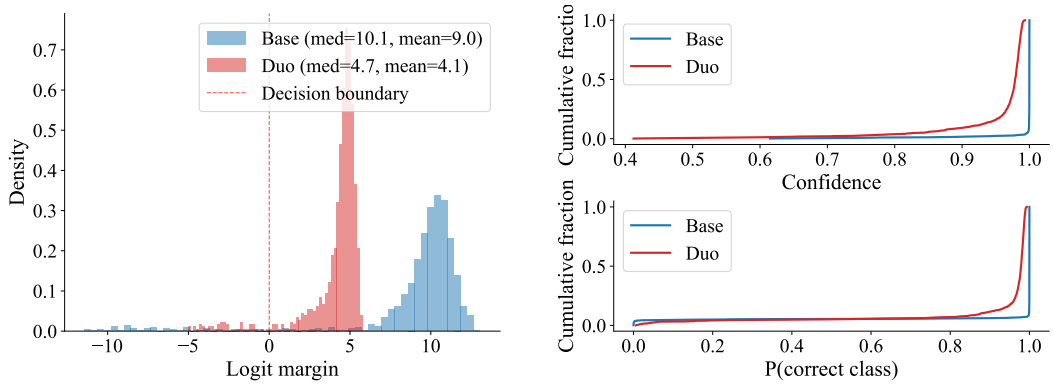


Figure 19: Logit margin (left) and empirical cumulative distribution functions (CDF) of probability and confidence (right) for Duo and Base for SciQ dataset.

B Full numerical results

B.1 Compatibility with calibration-aware methods

Table 4 reports the per-(metric, dataset) numbers that underlie Figure 2 of the main text. Within each B X vs Duo X pair, Duo X achieves the better NLL on every dataset across all three calibration families. On ECE, Duo X improves over B X in 17 of 18 cells, with the single exception of CSQA-BLoB where the two methods are essentially tied (B BLoB 2.04 vs Duo BLoB 2.06, within standard deviation). Accuracy is preserved or improved in 16 of 18 cells, with two minor exceptions where Base wins by a small margin (OBQA-BLoB and RACE-LA).

B.2 Learned Duo fusion weights

Table 5 reports the fusion weights (α, β) for every Duo variant in this paper, aggregated as mean $\pm$ standard deviation over 5 repetitions. Two regimes emerge clearly. For plain Duo and Duo LA the optimizer settles on $\alpha, \beta \ll 1$, consistent with both base and sidekick logits being sharply overconfident: the optimizer compresses them in addition to combining them, in line with the trivial-Duo ablation of Section 4.4. For Duo BLoB and Duo IB-EDL the calibration-aware base has already softened the answer-token logits, so the optimizer leaves them near unit scale ($\alpha \approx 1$) and tunes β to decide how much to incorporate the sidekick. On RACE and SciQ the BLoB sidekick is so saturated that β collapses to the optimizer's lower bound (10^{-6}), and Duo BLoB reduces to B BLoB on those datasets.

B.3 Mechanistic logit-space analysis of Asymmetric Duos

Table 6 reports the Base-Correct vs Base-Wrong decomposition of Section 5 for the remaining five benchmarks. The same asymmetry holds for every dataset: Duo trades a small confidence cost on Base's correct subset for a substantial reduction on Base's wrong subset.

C Detailed implementation

A reproducible codebase for all experiments of this work is provided at https://anonymous.4open.science/r/asymmetric-llm-duos-DC28.

Models. We fine-tune two models from the Qwen2 family [Yang et al., 2024b]: Qwen2-7B as the base model (f_{base}, B) and Qwen2-1.5B as the sidekick model (f_{side}, S). Both are loaded in `bfloat16`; Qwen2 has no dedicated padding token, so we reuse the end-of-sequence token `<|endoftext|>` for padding.

LoRA hyperparameters. We adapt both models with LoRA [Hu et al., 2022] via the PEFT library [Mangrulkar et al., 2022] and Transformers [Wolf et al., 2020]. LoRA adapters are applied to the four attention projections of every transformer block, `q_proj`, `k_proj`, `v_proj`, and `o_proj`. We use rank $r = 8$, scaling $\alpha_{\text{LoRA}} = 16$, dropout 0.1, and `bias = "lora_only"`, with `task_type = CAUSAL_LM`. The same configuration is shared across all six datasets and both model sizes.

Training details. For all datasets, we use the original train/validation/test splits provided by their respective authors, accessed via the Hugging Face datasets library [Wolf et al., 2020]; the sole exception is CSQA, which lacks a dedicated validation split, so we reserve 10% of its training data as the validation set. All MAP and IB-EDL runs use AdamW (HuggingFace `Trainer` defaults) with learning rate 5×10^{-5}, a cosine schedule with 20 warm-up steps, and `bfloat16` mixed precision. Per-device train and eval batch sizes are both 4. We train for 10 080 optimizer steps on every dataset except CSQA, which uses 30 000 steps to compensate for its smaller training split. The maximum tokenized prompt length is 300 tokens for ARC-C, ARC-E, CSQA, OBQA, and SciQ, and 512 tokens for RACE (whose passages are longer). For MAP we use weight decay 0; for IB-EDL we follow common evidential-training practice and add weight decay 10^{-3} together with gradient clipping at maximum global norm 20.0. We adopt the MCQA prompt templates and the next-token answer-classification setup of the IB-EDL reference implementation [Li et al., 2025] unchanged, so that differences between methods are attributable to the calibration procedure rather than to prompt design. Each method is run with 5 seeds and we report mean $\pm$ standard deviation across seeds. Validation and test answer-token logits at the final input position are saved once per run and reused by every post-hoc method below.

Computational details. All experiments were run on a DGX A100 cluster, using a single 40 GB GPU per run; we did not use multi-GPU or distributed training. Fine-tuning a base model on one dataset and evaluating on its test set takes roughly 30–40 minutes, with the exception of CSQA, which uses a larger number of training steps and takes around 90 minutes. For each dataset, we trained 5 base models and 5 sidekick models with

Table 4: Compatibility of the Duo procedure with calibration-aware baseline methods. Each block compares the calibration-aware base predictor B X against its Duo counterpart Duo X for $X \in \{\text{LA}, \text{BLoB}, \text{IB-EDL}\}$. Within each pair, the better entry per (metric, dataset) is in **bold**; ties are bolded on both rows. All metrics are mean $\pm$ SD over 5 repetitions. Plain-LoRA Duo and a Duo built from temperature-scaled Base and Sidekick are omitted because both reduce to the standard Duo of Table 1 (see remark in Section 4.3).

Metric	Method	ARC-C	ARC-E	CSQA	OBQA	RACE	SciQ
NLL ↓	B LA	$1.212_{\pm 0.052}$	$0.401_{\pm 0.029}$	$1.419_{\pm 0.096}$	$0.602_{\pm 0.068}$	$0.308_{\pm 0.006}$	$0.259_{\pm 0.016}$
	Duo LA	$\mathbf{0.438}_{\pm 0.006}$	$\mathbf{0.168}_{\pm 0.009}$	$\mathbf{0.432}_{\pm 0.010}$	$\mathbf{0.276}_{\pm 0.019}$	$\mathbf{0.255}_{\pm 0.005}$	$\mathbf{0.158}_{\pm 0.006}$
	B BLoB	$1.347_{\pm 0.101}$	$0.183_{\pm 0.039}$	$0.361_{\pm 0.008}$	$0.240_{\pm 0.016}$	$0.247_{\pm 0.009}$	$0.136_{\pm 0.002}$
	Duo BLoB	$\mathbf{0.443}_{\pm 0.014}$	$\mathbf{0.152}_{\pm 0.012}$	$\mathbf{0.355}_{\pm 0.007}$	$\mathbf{0.234}_{\pm 0.017}$	$\mathbf{0.245}_{\pm 0.009}$	$\mathbf{0.135}_{\pm 0.003}$
	B IB-EDL	$0.547_{\pm 0.018}$	$0.220_{\pm 0.021}$	$0.556_{\pm 0.017}$	$0.368_{\pm 0.016}$	$0.267_{\pm 0.003}$	$0.205_{\pm 0.005}$
	Duo IB-EDL	$\mathbf{0.509}_{\pm 0.015}$	$\mathbf{0.183}_{\pm 0.018}$	$\mathbf{0.487}_{\pm 0.018}$	$\mathbf{0.310}_{\pm 0.016}$	$\mathbf{0.257}_{\pm 0.003}$	$\mathbf{0.182}_{\pm 0.005}$
Acc ↑	B LA	$86.66_{\pm 0.48}$	$95.11_{\pm 0.32}$	$85.32_{\pm 0.42}$	$90.04_{\pm 0.14}$	$\mathbf{91.80}_{\pm 0.15}$	$94.84_{\pm 0.23}$
	Duo LA	$\mathbf{87.12}_{\pm 0.32}$	$\mathbf{95.23}_{\pm 0.40}$	$\mathbf{85.81}_{\pm 0.67}$	$\mathbf{90.64}_{\pm 0.17}$	$91.75_{\pm 0.49}$	$\mathbf{95.22}_{\pm 0.33}$
	B BLoB	$86.31_{\pm 0.67}$	$94.89_{\pm 0.38}$	$86.55_{\pm 0.61}$	$\mathbf{91.84}_{\pm 0.83}$	$\mathbf{91.57}_{\pm 0.59}$	$95.42_{\pm 0.08}$
	Duo BLoB	$\mathbf{86.55}_{\pm 0.30}$	$\mathbf{94.97}_{\pm 0.50}$	$\mathbf{86.58}_{\pm 0.62}$	$91.36_{\pm 0.83}$	$\mathbf{91.57}_{\pm 0.59}$	$\mathbf{95.44}_{\pm 0.11}$
	B IB-EDL	$86.28_{\pm 0.60}$	$95.23_{\pm 0.17}$	$85.39_{\pm 0.52}$	$90.84_{\pm 0.38}$	$91.85_{\pm 0.26}$	$95.02_{\pm 0.22}$
	Duo IB-EDL	$\mathbf{86.55}_{\pm 0.67}$	$\mathbf{95.45}_{\pm 0.24}$	$\mathbf{85.57}_{\pm 0.76}$	$\mathbf{90.92}_{\pm 0.59}$	$\mathbf{91.88}_{\pm 0.26}$	$\mathbf{95.24}_{\pm 0.32}$
ECE ↓	B LA	$9.79_{\pm 0.39}$	$3.55_{\pm 0.31}$	$10.95_{\pm 0.72}$	$7.03_{\pm 0.42}$	$4.77_{\pm 0.23}$	$3.81_{\pm 0.25}$
	Duo LA	$\mathbf{2.02}_{\pm 0.41}$	$\mathbf{1.08}_{\pm 0.43}$	$\mathbf{2.37}_{\pm 0.76}$	$\mathbf{2.84}_{\pm 0.37}$	$\mathbf{2.12}_{\pm 0.25}$	$\mathbf{1.71}_{\pm 0.23}$
	B BLoB	$12.27_{\pm 0.67}$	$5.48_{\pm 2.90}$	$\mathbf{2.04}_{\pm 0.15}$	$2.71_{\pm 0.75}$	$2.17_{\pm 0.31}$	$1.55_{\pm 0.38}$
	Duo BLoB	$\mathbf{4.24}_{\pm 0.47}$	$\mathbf{0.97}_{\pm 0.21}$	$2.06_{\pm 0.30}$	$\mathbf{2.64}_{\pm 0.52}$	$\mathbf{1.48}_{\pm 0.35}$	$\mathbf{1.39}_{\pm 0.24}$
	B IB-EDL	$4.88_{\pm 1.04}$	$3.90_{\pm 1.01}$	$5.61_{\pm 0.78}$	$9.37_{\pm 1.20}$	$3.84_{\pm 0.55}$	$2.89_{\pm 0.65}$
	Duo IB-EDL	$\mathbf{2.73}_{\pm 1.03}$	$\mathbf{1.07}_{\pm 0.14}$	$\mathbf{3.65}_{\pm 0.52}$	$\mathbf{2.81}_{\pm 0.68}$	$\mathbf{2.10}_{\pm 0.64}$	$\mathbf{2.19}_{\pm 0.41}$

Table 5: Learned Duo fusion weights (α, β) on the validation logits of base and sidekick, mean $\pm$ SD over 5 repetitions. Values reported as 0.000 for β are at the optimizer's lower bound (10^{-6}).

Method	Weight	ARC-C	ARC-E	CSQA	OBQA	RACE	SciQ
Duo	α	$0.162_{\pm 0.006}$	$0.220_{\pm 0.012}$	$0.120_{\pm 0.008}$	$0.219_{\pm 0.022}$	$0.568_{\pm 0.022}$	$0.330_{\pm 0.013}$
	β	$0.046_{\pm 0.010}$	$0.072_{\pm 0.012}$	$0.071_{\pm 0.005}$	$0.126_{\pm 0.017}$	$0.083_{\pm 0.017}$	$0.135_{\pm 0.020}$
Duo LA	α	$0.227_{\pm 0.012}$	$0.289_{\pm 0.015}$	$0.167_{\pm 0.008}$	$0.281_{\pm 0.015}$	$0.626_{\pm 0.027}$	$0.399_{\pm 0.014}$
	β	$0.068_{\pm 0.012}$	$0.080_{\pm 0.015}$	$0.100_{\pm 0.005}$	$0.155_{\pm 0.029}$	$0.056_{\pm 0.022}$	$0.111_{\pm 0.023}$
Duo BLoB	α	$0.228_{\pm 0.013}$	$1.540_{\pm 0.118}$	$0.922_{\pm 0.041}$	$1.022_{\pm 0.097}$	$1.116_{\pm 0.010}$	$1.031_{\pm 0.032}$
	β	$0.049_{\pm 0.014}$	$0.142_{\pm 0.099}$	$0.261_{\pm 0.028}$	$0.145_{\pm 0.057}$	$0.000_{\pm 0.000}$	$0.001_{\pm 0.003}$
Duo IB-EDL	α	$0.960_{\pm 0.053}$	$0.936_{\pm 0.095}$	$0.865_{\pm 0.030}$	$0.981_{\pm 0.045}$	$1.062_{\pm 0.022}$	$0.820_{\pm 0.023}$
	β	$0.411_{\pm 0.054}$	$0.430_{\pm 0.081}$	$0.484_{\pm 0.023}$	$0.428_{\pm 0.066}$	$0.135_{\pm 0.023}$	$0.320_{\pm 0.022}$

different random seeds and report the mean and standard deviation across these runs. Computing the Duo fusion weights on top of the fine-tuned models takes only seconds.

Asymmetric Duo fusion. The fusion scales (α, β) are estimated on the validation logits with L-BFGS-B, initialized at $(\alpha, \beta) = (1.0, 1.0)$ and bounded below by 10^{-6} on each coordinate (no upper bound, no sum constraint). The loss is the validation NLL of $\mathrm{softmax}(\alpha z_{\text{base}} + \beta z_{\text{side}})$, evaluated in float64 with the standard log-sum-exp stabilization. The same optimizer settings are used for every Duo variant in the paper, including the calibration-aware Duos (Duo LA, Duo BLoB, Duo IB-EDL), where α and β are fit on the validation logits of the corresponding calibration-aware Base and Sidekick. Logits from the two models are matched by example index before fitting.

Temperature scaling. A single scalar temperature $T > 0$ per model is fit on the validation NLL via L-BFGS-B in log-space (initialized at $T = 1$) with the analytical gradient $\partial\,\mathrm{NLL}/\partial \log T = (1/T)\,\mathbb{E}_i[\,\ell_{i,y_i} - \mathbb{E}_{p_i}[\ell_i]\,]$, then applied to the test logits [Guo et al., 2017]. We optimize on float64 logits to avoid the gradient collapsing to zero under float32 precision.

Table 6: NLL and ECE of Base and Duo, computed separately on test set samples of benchmark datasets that Base classifies correctly/incorrectly. Bold indicates the better value between Base and Duo within each subset.

Metric	Dataset	Method	Base Correct	Base Wrong
NLL ↓	ARC-C	Base	**0.001**	16.426
		Duo	0.090	**2.996**
	ARC-E	Base	**0.001**	13.313
		Duo	0.041	**2.935**
	OBQA	Base	**0.001**	11.055
		Duo	0.087	**2.224**
	RACE	Base	**0.016**	4.041
		Duo	0.055	**2.638**
	SciQ	Base	**0.003**	6.625
		Duo	0.037	**2.702**
ECE ↓	ARC-C	Base	**0.08**	97.53
		Duo	8.13	**78.04**
	ARC-E	Base	**0.06**	98.40
		Duo	3.47	**76.51**
	OBQA	Base	**0.11**	99.50
		Duo	6.88	**61.39**
	RACE	Base	**1.30**	82.71
		Duo	4.66	**72.57**
	SciQ	Base	**0.25**	95.65
		Duo	3.45	**74.08**

Laplace-LoRA. We use the official `bayesian-lora` package [Yang et al., 2024a]. Kronecker factors are computed on the training loader with at most $n_{\text{KFAC}} = 10$ low-rank components per factor and a low-rank threshold of 100. The prior variance σ^2 is initialized at 0.1 and tuned by maximizing the model evidence [Yang et al., 2024a]. Predictive variances are produced via the Jacobian-mean approximation. We did not perform additional hyperparameter tuning beyond the package defaults.

IB-EDL implementation details. We follow the IB-EDL formulation and implementation of Li et al. [2025]. The pre-evidence head is the language-model head of the LLM evaluated at the last input position and restricted to the answer-token columns; positivity is enforced with a softplus and a Dirichlet over the K answer choices is constructed by adding 1 to the evidence vector. The variational information bottleneck draws $K_{\text{IB}} = 20$ Gaussian noise samples per training example, keeps $\sigma_{\text{mult}} = 0$ at evaluation (i.e. no post-hoc evidence shrinkage), and uses the sum-of-squares Bayes-risk loss without label smoothing. The KL regularization weight is ramped up linearly starting at 10% of training; we set the final regularization weight to 0 since the IB term itself already regularizes the evidence. The information-bottleneck coefficient β is the only IB-EDL hyperparameter we tune—per (model, dataset) pair—using the small grid in Table 7 in Appendix C.1.

BLoB. BLoB [Wang et al., 2024] is run with the authors' official `bayesian-peft` codebase using its default settings, including 10 Monte-Carlo samples at evaluation. We average the per-sample probabilities to form the Bayesian predictive and convert back to log-probabilities so that the saved BLoB logits live in the same log-probability space as the other methods.

Deep ensemble baseline (B DE). For each dataset we form 5 random ensembles of size 2 from the 5 available training seeds. Predictions are averaged in probability space, and metrics are then computed on each sampled pair and averaged over pairs.

Calibration metrics. ECE is computed with 15 equal-width confidence bins. NLL is the mean $-\log p_y$ over the test set, with a 10^{-12} floor on probabilities for numerical stability.

C.1 Selection of the IB-EDL β

The IB-EDL objective is governed by an information-bottleneck coefficient β that weights the KL regularizer relative to the evidential likelihood [Li et al., 2025]. The optimal β does not transfer across model scales or datasets and must be selected from a small grid per (model, dataset) pair. For each dataset, we evaluated 4–5

candidate values spanning roughly two orders of magnitude (Table 7), and independently for Base and Sidekick, we picked the candidate that minimized the hold-out ECE on seed 1; the selected entries are bold-faced. The chosen β values were then frozen across all 5 repetitions used to produce the IB-EDL rows of Tables 1 and 4.

Table 7: Hold-out ECE (seed 1) of the base and sidekick models across the candidate β values evaluated for each dataset. The bold entry in each (dataset, model) column is the value used in the main experiments.

Dataset	β	Base ECE	Sidekick ECE
ARC-C	9×10^{-5}	**0.0439**	0.1118
	1.5×10^{-4}	0.0467	0.0986
	5×10^{-4}	0.1207	**0.0538**
	1.0×10^{-3}	0.1633	0.0898
ARC-E	2×10^{-6}	**0.0424**	0.0548
	2×10^{-5}	0.0684	**0.0508**
	9×10^{-5}	0.1138	0.0693
	1.0×10^{-4}	0.1065	0.0796
CSQA	5×10^{-5}	0.0577	0.0638
	9×10^{-5}	**0.0501**	**0.0417**
	1.5×10^{-4}	0.0686	0.0505
	5×10^{-4}	0.1224	0.0968
	1.0×10^{-3}	0.1515	0.1283
OBQA	6×10^{-5}	0.0812	**0.0560**
	9×10^{-5}	**0.0811**	0.1150
	1.5×10^{-4}	0.0880	0.0710
	5×10^{-4}	0.1357	0.1122
	1.0×10^{-3}	0.1606	0.1283
RACE	1×10^{-6}	0.0425	0.0477
	2×10^{-6}	**0.0405**	0.0476
	7×10^{-6}	0.0526	**0.0461**
	9×10^{-5}	0.0976	0.0791
SciQ	5×10^{-7}	0.0300	**0.0386**
	1×10^{-6}	**0.0179**	0.0493
	2×10^{-6}	0.0400	0.0476
	9×10^{-5}	0.1082	0.0924

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